Biological samples collection information for age 53 years (1999)

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Document details

Categories of variables	Cardiovascular health
Summary of work undertaken	Cleaning of Non-fasting blood measures: HBA1C99D =glycated haemoglobin (%) HB99D = Haemoglobin CHOLB99D =Total cholesterol (mmol/l) TRIG99D =triglycerides (mmol/l) HDLC99D =high density lipoprotein cholesterol (mmol/l) LDLC99D = low density lipoprotein cholesterol (mmol/l)
Names of source variables	HBA1C99D, HB99D, CHOLB99D, TRIG99D, HDLC99D, LDLC99D
Output variables	chol99n, trig99n, hdl99n, ldl99n, hba99n
Papers using these variables	Skidmore, P.M.L., Hardy, R.J., Kuh, D.J., Langenberg, C., Wadsworth, M.E.J. (2004) Birth weight and lipids in a national birth cohort study. Arteriosclerosis, Thrombosis and Vascular Biology, 24, 588-594.

Data collection methods

Data collection was carried out at home visits to cohort members by research nurses recruited and employed by the National Centre for Social Research. An electronic questionnaire was administered by nurses using laptop computers. Nurses took a buccal swab from which DNA was derived. Nurses also took non- fasting blood samples for DNA, cardiovascular risk factors (cholesterol, HDL cholesterol, triglyceride, glycosolated haemoglobin), [a haemoglobin, red and white cell counts] and nutritional markers (red cell folate, plasma folate, plasma B12, plasma ferritin). Samples were posted to laboratories for analysis. The cholesterol, HDL cholesterol, triglyceride, and glycosylated haemoglobin were measured at the Royal Victoria Hospital in Newcastle England in 2000. Total cholesterol was measured by enzymatic CHOD-PAP. Precipitation for measurement of HDL cholesterol was performed using phosphotungstic Mg 2+ ; triglycerides were measured using a glycerol/kinase POD-linked reaction of glycerol liberated enzymatically from triglycerides. All of these measurements were made with a Bayer DAX-72. LDL cholesterol was calculated using the Friedwald formula; LDL cholesterol= total cholesterol - HDL cholesterol - 0.45 x triglycerides. HbA1c was measured using an ion exchange method on a Tosoh A1c 2.2 analyser

The blood counts (haemoglobin, red and white cell counts) and nutritional markers (red cell folate, plasma folate, plasma B12, plasma ferritin) were done at RVI. The methods used were as follows: Hb – standard spectrophotometric analysis on a Coulter Electronics STKS analyzer.Red Cell folate / plasma folate /B12 and ferritin – all measured using specific immunoassays on an Abbott Imx analyzer. The residual blood went to Dallas for DNA, and she logged the identifiers. She transferred blood to blood transfusion that did blood typing and sent the residual to ECACC (according to Warren Hilder). Dallas says no blood went to ICH in 1999. She has forwarded query to Lynne and I await the result. The buccal samples also were sent to UCL (Prof Dallas Swallow at UCL). Full details are in the nurse manual at I:\LHA\Questionnaires.

Raw variables

HBA1C99D =glycated haemoglobin (%) CHOLB99D =Total cholesterol (mmol/l) TRIG99D =triglycerides (mmol/l) HDLC99D =high density lipoprotein cholesterol (mmol/l) LDLC99D = low density lipoprotein cholesterol (mmol/l)

Paula Skidmore looked into how cholesterol was measured – from birth weight and lipids paper:

Total cholesterol was measured by enzymatic CHOD – PAP. Precipitation for measurement of high density lipoprotein (HDL) cholesterol was carried out using phosphotungstic Mg2+, triglycerides were measured using a glycerol/kinase POD linked reaction of glycerol liberated enzymatically from triglycerides. All of these measurements were made with a Bayer DAX-72. Low density lipoprotein (LDL) cholesterol was calculated using the Friedwald formula:

LDL cholesterol (mmol/l) = total cholesterol - HDL cholesterol - 0.45 triglycerides.

Cleaned by Claudia L.

- Set not interviewed to 10000
- Set these ranges as valid values and rest to SYSMIS
- HBA1C99D 4.0 - 13.8
- HB99D 7.9 - 18.7
- CHOLB99D 2.4 - 12.8
- TRIG99D 0.3 - 16.5
- HDLC99D 0.2 - 8.0 Note: HDL values of 8 are excluded (as discussed with MP) and coded as 98. These are SERNO: 027011, 126068 and 134019
- LDLC99D 0.6 - 8.0 Note: LDL values of 8 are also excluded now and coded as 98

As these then do not add up to 5362 and lose information about missingness, I have redefined CLs variables within the job Missing values are then 77, 98, 99=no interview

Cleaned variable names:

- HBA99N
- CHOL99N
- TRIG99N
- HDL99N
- LDL99N

Results

Following results files were received.

- RVI sent us the results in excel spreads sheets.
- RVI gave people SEPR numbers (SSN numbers).
- Data came back in 3-4 data files with SSN numbers which were converted into SPSS files with our serial numbers in them. These SPSS files were nor converted into a library file.
- Warren has a series of project files
- SSN numbers are also in CAPI 1999 list of projects
- There may be one called NRVI 1999. These should be in black binders

Look at who used the data. The original data files prepared for them may give us a clue as to where the variables are. Potential names, David Greenberg, Celia Greenberg, Anna Rickard, DUNN nutrition unit, HNR.

Derivations

HB99N

Purpose: Haemoglobin at age 53yrs Year: 1999

Definitions

77 original missing value (no blood taken?)
98 excluded values 99 Not interviewed

Specification

RECODE HB99D (7.9 THRU 18.7=COPY) (18.7 THRU 70=98) (ELSE=COPY) INTO HB99N

CHOL99N

Purpose: Total cholesterol (mmol/l) at age 53yrs Year: 1999

Definitions

77 original missing value (no blood taken?)
98 excluded values 99 Not interviewed

Specification

RECODE CHOLB99D (2.4 THRU 12.8=COPY) (12.8 THRU 70=98) (ELSE=COPY) INTO CHOL99N

TRIG99N

Purpose: Triglycerides (mmol/l) at age 53yrs Year: 1999 Year: 1999

Definitions

77 original missing value (no blood taken?)
98 excluded values 99 Not interviewed

Specification

RECODE TRIG99D (0.3 THRU 16.5=COPY) (16.5 THRU 70=98) (ELSE=COPY) INTO TRIG99N

HDL99N

Purpose: High density lipoprotein cholesterol (mmol/l) at age **53yrs** Year: 1999

Definitions

77 original missing value (no blood taken?)
98 excluded values 99 Not interviewed

Specification

RECODE HDLC99D (0.2 THRU 8.0=COPY) (8.0 THRU 70=98) (ELSE=COPY) INTO HDL99N

LDL99N

Purpose: Low density lipoprotein cholesterol (mmol/l) at age **53yrs** Year: 1999

Definitions

77 original missing value (no blood taken?)
98 excluded values 99 Not interviewed

Specification

RECODE LDLC99D (0.6 THRU 8.0=COPY) (8.0 THRU 70=98) (ELSE=COPY) INTO LDL99N

References

Claudia Langenberg, MD, , PhD, Michael E. J. Wadsworth, PhD, Eric Brunner, PhD, and , PhD.Social Circumstances and Education: Life Course Origins of Social Inequalities in Metabolic Risk in a

Prospective National Birth Cohort. Am J Public Health. 2006; 96:2216–2221.

Variables in this document

5 high-confidence matches

Variable	Label	How it was found
Biological samples [52] › Blood assays [201] › Blood biochemistry [2011]		
chol99n	Cholesterol (mmol/l) - at age 53 years	topsheet field
hba99n	Glycohaemoglobin - HbA1c (%) - at age 53 years	topsheet field
hdl99n	High-density lipoprotein - HDL (mmol/L) at age 53 years	topsheet field
ldl99n	Low-density lipoproteins - LDL (mmol/L) at age 53 years	topsheet field
trig99n	Triglcyeride (mmol/l) - at age 53 years	topsheet field

2 medium-confidence matches

Variable	Label	How it was found
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Other medical conditions [51011]		
blood	Have you had anaemia in the last 10 years - at age 43 years	prose scan
Questionnaire data [55] › Physical activity and exercise [521] › EPIC Physical activity questionnaire [5211] › EPIC Physical activity questionnaire - derived variables [52111]		
employed	Reported currently employed/volunteering and/or gave work activities (Q2) - at age 60-64 years	prose scan

3 low-confidence matches

Variable	Label	How it was found
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Musculoskeletal system conditions [51016]		
back	In the last 12 months, have you had sciatica, lumbago or severe backache at age 53 years	prose scan
Questionnaire data [55] › Sociodemographics [515] › Marital status [5153]		
mary	Thinking of your first marriage, when were you married. Can you tell me the year? - at age 53 years	prose scan
Specialist questionnaire data [56] › Womens health questionnaire [700] › Hysterectomy [7006]		
sent	Was the participant sent a hysterectomy questionnaire in 2005? - at age 59 years	prose scan